

RESEARCH ARTICLE

Mesoscopic vascular alterations in Alzheimer's disease

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Abstract

INTRODUCTION: Cerebrovascular alterations are associated with the pathology of Alzheimer's disease (AD). Yet, the role of these alterations is not fully understood, partly due to a paucity of data from mesoscopic vasculature (24–240 µm diameter).

METHODS: We used label-free, serial-sectioning optical coherence tomography to reconstruct mesoscopic cerebrovasculature of the dorsolateral prefrontal cortex (DLPFC) samples from AD and control subjects. We quantified three-dimensional alterations to the vascular networks and measured the correlations between vascular alterations and deposits of amyloid β protein ($A\beta$) and phosphorylated tau protein (p-tau).

RESULTS: The AD group had significantly reduced volume fraction, vessel length density, and branch density. There were negative trends between a subset of the vascular metrics and the density of $A\beta$ and p-tau.

DISCUSSION: AD samples in DLPFC present with significant mesoscopic cerebrovasculature alterations. This insight helps close the knowledge gap between micro- and macroscopic cerebrovascular pathologies.

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KEYWORDS

Alzheimer's disease, amyloid beta, cerebrovascular network, histopathology, human brain, optical coherence tomography, phosphorylated tau, vascular morphology

Highlights

- Volumetric imaging of mesoscopic cerebrovasculature with serial-sectioning optical coherence tomography (OCT).
- Vascular metrics: volume fraction, length density, branch density, and tortuosity.
- Human ex vivo samples from dorsolateral prefrontal cortex (Alzheimer's disease [AD] and controls).
- Compared to controls, AD had reduced vascularity.
- Serial-sectioning OCT provides novel insights into mesoscopic cerebrovasculature.

1 | INTRODUCTION

With an aging global population, the prevalence of Alzheimer's disease (AD) is forecasted to reach 1 in 85 by the year 2050.^{1,2} AD ultimately causes memory and cognitive impairment, autonomic dysfunction, and potentially fatal complications.³ Given that there is currently no cure for AD, it is imperative that the pathogenesis be fully elucidated to better inform the development of effective therapeutics.

The traditional histopathological hallmarks of AD are extracellular amyloid β (A β) protein plaques and intracellular phosphorylated tau (p-tau) neurofibrillary tangles (NFTs) within neurons.⁴ These plaques and tangles were originally believed to drive the pathogenesis.⁵ Subsequent studies have identified cerebrovascular abnormalities in AD, leading to the hypothesis of vascular-driven pathologies.^{5,6} Overall, there is an incomplete understanding of the precise role of vascular dysfunction in AD.⁷ This is due in part to limitations of the current tools for visualizing comprehensive vasculature in *post mortem* human brains, which have constrained researchers to focus primarily on the capillary beds, inadvertently neglecting the mesoscopic arterioles and venules.

Our laboratory has demonstrated a mesoscopic, label-free, three-dimensional vascular imaging system⁸ to quantify alterations in mesoscopic cerebrovasculature (defined in this study as blood vessels with a diameter ranging from 24 to 240 μ m).⁹ This system consists of a swept-source optical coherence tomography (OCT) microscope, a three-axis translation stage, and an integrated vibratome. OCT microscopes measure axial discontinuities in the refractive indices of tissue and the differences in scattering coefficients (A-scans). In *post mortem* tissue samples, the blood vessel lumens are devoid of light-scattering particles, which creates a dark contrast with respect to the surrounding tissue medium, enabling segmentation of the vasculature. The three-axis translation stage facilitates a raster of A-scans across the x- and y-axes, generating a volumetric image of the surface layer of tissue within the imaging depth (C-scan). In our custom system, a three-axis translation stage moves the block-face tissue, and an integrated vibratome sections off the surface layer of tissue. Then, the imag-

ing is repeated with the newly exposed surface. These imaging and slicing sequences are repeated throughout the entire depth of the tissue, motivating the system's name—serial-sectioning OCT (sOCT). One critical feature of sOCT is that imaging is acquired from the tissue block-face prior to sectioning, thus minimizing tissue distortion, which is unavoidable in histology. The vascular images from sOCT have been validated using tomato lectin staining and confocal imaging.⁸

In this study, we used sOCT to image, reconstruct, and characterize the mesoscopic vasculature in *post mortem* human brain tissue from late-stage AD patients and a group of age-matched controls. We examined the vascular changes in the dorsolateral prefrontal cortex (DLPFC), a brain region involved in high-level executive functions that suffers impairments and accumulates A β plaques and NFTs in AD.¹⁰ Automated software tools were developed to extract the vascular networks and quantify their morphological metrics. This enabled the identification of significant volumetric alterations to the mesoscopic vasculature within the DLPFC in late-stage AD.

2 | METHODS

2.1 | Tissue samples

We imaged 13 anonymized human DLPFC brain tissue samples: five AD and eight age-matched controls. Each tissue volume was approximately 2.0 $\times$ 2.0 $\times$ 0.35 cm. The AD samples were sourced from Boston University's Alzheimer's Disease Research Center (ADRC), and the control samples were sourced from the Understanding Neurological Injury and Traumatic Encephalopathy (UNITE)¹¹ Brain Bank (formerly the VA-BU-CLF Brain Bank).

The AD staging followed both the Braak¹² and Thal¹³ systems. The AD cohort consisted entirely of late-stage AD: Braak VI (B3), Thal 4 (A3)–Thal 5 (A3), and CERAD C2–C3. Each AD subject in this study had a comorbidity of CAA, staged by the Vonsattel Grading System.¹⁴ The inclusion criterion for the AD group was late-stage AD, and the inclusion criteria for the control group were the absence of clinically

RESEARCH IN CONTEXT

1. **Systematic review:** The authors used both traditional and artificial intelligence search tools (PubMed and Consensus, respectively). The literature on cerebrovascular alterations in neurodegeneration focuses primarily on capillaries, and we appropriately cited the relevant studies. In comparison, studies of changes to cortical arteries focused on arterial wall thickening and stiffening, which were not the focus of this study. Overall, there was a paucity of literature examining the mesoscopic cerebrovasculature in neurodegenerative disorders.
2. **Interpretation:** We quantified the morphological metrics of mesoscopic, volumetric cerebrovascular networks from *post mortem* human brain tissue. We identified significantly reduced vascularity in late-stage Alzheimer's disease compared to controls.
3. **Future directions:** Our work demonstrates a robust method for quantifying mesoscopic vasculature in the human cortex. We plan to strengthen the outcomes by expanding the study to a larger cohort and incorporating additional pathologies. Furthermore, we will investigate the relationship between disease staging and mesoscopic vascular alterations.

diagnosed neurodegenerative disorders and an age at death approximately equivalent to the AD group. The exclusion criteria for both groups were any of the following: frontotemporal lobar degeneration, amyotrophic lateral sclerosis, and neocortical Lewy body disease. The AD and control groups both had arteriosclerosis staged by a neuropathologist and were approximately matched for arteriosclerosis severity (Table S1). The *post mortem* interval (PMI) was below 24 h for all samples except two controls (43.5 and 51.8 h).

After imaging, two of the control samples were excluded due to low imaging contrast, which would have impacted the vascular network reconstruction. Since aging results in cardiovascular deterioration^{15,16} and cerebrovascular degeneration,^{17,18} a large age range can introduce a moderating factor. Therefore, an additional control sample was excluded to better match the age of the experimental group. In total, this study included five AD samples (two females and three males) with an average age of 78.8 ± 8.6 years and five age-matched control subjects (all male) with an average age of 75.8 ± 8.3 years.

Both brain banks preserved the tissue samples in a fixative agent (periodate-lysine-paraformaldehyde) and then refrigerated them at 4°C for a minimum of 2 months. Prior to imaging, the tissue samples underwent a 1-month washing procedure to remove the fixative. This involved submerging each sample in a solution of 0.01 M phosphate-buffered saline (PBS) and then placing the sample on an orbital shaker, stored at 4°C. Next, a custom three-dimensional mold was used to embed each tissue sample in an agarose gel (4.5% by mass). After

embedding, the samples were submerged in a preservative solution (0.01 M PBS with 0.1% sodium azide) and stored at 4°C.

2.2 | sOCT image acquisition

The sOCT imaging system was designed and built in-house and has been previously described.^{8,19} The system consists of a 4x Olympus objective (UPLFLN4x, NA = 0.13) and a swept-source laser (Axsun Technologies, Billerica, MA, USA) with a central wavelength of 1310 nm and a bandwidth of 135 nm. As implemented, the system has axial and lateral resolutions of 5.6 and 6 μ m, respectively. The field of view of each imaged tile is 3 $\times$ 3 mm with a 15% overlap between adjacent imaging tiles.

The tissue samples were mounted in a bath containing 0.01 M PBS and fastened to a three-axis translation stage, which facilitated image acquisition of the entire tissue surface. For each acquisition, only the top 150 μ m of the tissue volume was used for volumetric reconstruction and stitching. After imaging the entire surface of the tissue volume, the translation stages would move the mounted tissue volume along the integrated vibratome to section off the top 150 μ m of tissue into a variable number of sections. Each of these sections ranged in thickness from 30 to 150 μ m, which facilitated a range of histopathological assessments.

Each 150- μ m acquisition was reconstructed by stitching together the tiles with an algorithm that identified the globally optimal configuration of a collection of three-dimensional microscopy images.²⁰ This procedure was repeated for each 150- μ m depth, resulting in a stack of C-scans that reconstructed the entire depth of the tissue volume (Figure 1A). The reconstructed tissue volumes were downsampled, resulting in a voxel dimension of $12 \times 12 \times 15 \mu$ m (x, y, and z).

2.3 | Data analysis

The resulting image volumes contained a decay along each A-scan depth, reflecting the structure-dependent scattering coefficient (Figure 1A). This resulted in a periodic artifact in the signal intensity of the reconstructed tissue volume. To correct this, we normalized each tissue depth in a two-step process. First, we convolved each depth with a two-dimensional box linear filter for each depth. Second, we divided the original OCT intensity by the output of this convolution at each depth. The result was a uniform intensity along the entire depth of the volume (Figure 1B), only highlighting fine structures like vessels.

A Hessian-based vessel enhancement filter (Frangi filter) was applied to each normalized volume to segment the vasculature.²¹ The Frangi filter first applies a Hessian matrix to the intensity volume. Then, it enhances tubular structures by analyzing the resulting eigenvalues to identify the minimal curvature in a local neighborhood, which corresponds to the principal direction of blood vessels. The Frangi filter outputs a probability for each voxel, where 1 represents a 100% probability of the voxel belonging to a vessel. We applied a probability threshold such that the voxels with a probability below the threshold

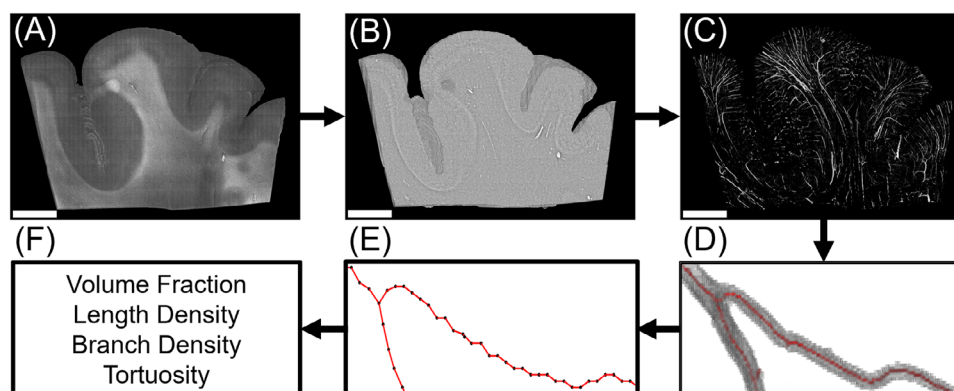


FIGURE 1 Process of extracting the three-dimensional vascular network from the OCT volume from a control case. (A) The reconstructed and registered OCT C-scan stacks. (B) The same OCT C-scan stacks from (A) after normalizing to remove the depth intensity decay artifact and eroding the borders to remove false positives along the edges. (C) The segmented three-dimensional vascular network from (B). (D) A subset of segmentation (gray) circumscribing the center line (red). (E) The graphical representation of the center line (nodes are black and edges are red). (F) A list of the calculated vascular metrics from the graph. The scale bars in (A), (B), and (C) all represent 5 mm. OCT, optical coherence tomography.

were converted to zero and those above were converted to one. For these data, the probability threshold was 0.18, which was the empirically determined optimum in a previous study of *post mortem* brain tissue.⁸ The result was a binary matrix, where each non-zero voxel represented a portion of a vessel. These binary segmentation matrices contained false positives from pia and agarose along the borders of the tissue volumes. To remove these false positives, we applied a tissue mask that retained vascular segmentation, resulting in a reconstruction of the vascular network (Figure 1C and Video S1).

Next, we utilized a medial axis thinning algorithm to identify the center line of the vasculature (Figure 1D).^{22,23} The center lines were then converted into a graphical representation of nodes and edges (Figure 1E).²⁴ The graph facilitated mathematical analyses that would otherwise not be possible with the segmentation alone. The graphical representations were used to compute the following vascular metrics: volume fraction, length density, branch density, and tortuosity (Figure 1F).

The volume fraction is a unitless measure defined as the total number of voxels belonging to vasculature divided by the total number of voxels belonging to tissue and vasculature. The length density measures the total length of blood vessels within the tissue volume divided by the total volume of tissue containing them (mm/mm^3). Similarly, the branch density is the total number of branch points within a tissue volume divided by the total tissue volume ($1/\text{mm}^3$). Tortuosity is a unitless measure of how much each blood vessel curves, and it is defined as the ratio of the total arc length to the Euclidean distance between the two endpoints.

The vascular metrics were measured across the entirety of each tissue volume and the following subdivisions: white matter (WM), gray matter (GM), sulci, gyri, sulcal WM, sulcal GM, gyral WM, and gyral GM. The gyri and sulci subdivisions sampled the gyral crowns and the sulcal ends (Figure S1). The gyri and sulci masks were manually annotated in Fiji.²⁵ They spanned approximately 4 mm across the y-axis of the image space (superior to inferior with respect to anatomical orientation). Each mask contained roughly equal proportions of WM and GM.

The gyri masks extended 4 mm inferior to the peaks of the gyri. In comparison, the sulci masks extended approximately 1 mm superior and 3 mm inferior to the sulci trough border (defined here as the boundary between the pia matter and the WM in the sulci trough). Masks for the subdivisions were manually generated by annotating the sOCT intensity volume in Fiji. The subdivision masks were then applied to the respective vessel segmentation networks for subsequent analyses.

Additionally, each tissue volume was digitally sampled with a uniform grid of 1 mm^3 regions of interest (ROI). As the ROIs were disjoint, each contained a disjoint subset of the vascular segmentation from the entirety of the volume. The segmentation within each ROI was converted into a graph, and then the vascular metrics (volume fraction, length density, branch density, and tortuosity) were calculated separately for each ROI. This resulted in an array of four vascular measurements for each ROI.

Next, to visually examine the distribution of the vascular metrics, each tissue sample was visualized as a two-dimensional heatmap. For each vascular metric, the respective ROIs were averaged across the z-dimension, excluding ROIs that lay outside of the tissue mask. Then, we performed a maximum intensity projection (MIP) across all depths of the tissue masks. The resulting MIP was used as the tissue mask for the two-dimensional heatmap. This process was performed separately for each vascular metric.

2.4 | Quantitative histopathology

Each tissue sample underwent histopathological evaluation for A β and p-tau. Since the sOCT vasculature signal had already been validated against a lectin stain, an additional vascular stain was not performed. Instead, we focused on the primary pathologies of AD. Two 30 μm -thick sections were randomly selected from each tissue volume after sOCT imaging. Each section was mounted to a charged slide and submerged in -20°C acetone for 30 min. The slides were then rinsed in tap water for 5 min and placed into a PBS solution

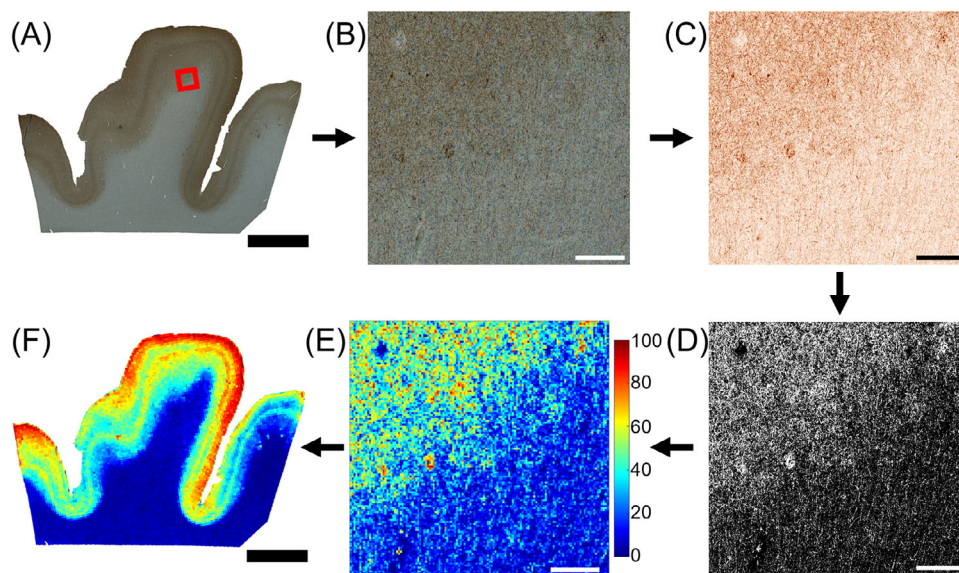


FIGURE 2 The process of segmenting the histopathology stain and generating a heatmap. (A) The brightfield image of an AD sample stained with AT8 antibody. (B) The full-resolution cropped region from the red highlighted region in (A). (C) The output of color deconvolution within the region in (B), resulting in just the AT8 stain. (D) The output of applying a minimum threshold to (C), segmenting the stain (white) from the parenchymal tissue (black). (E) A representative heatmap from the region in (D) (units are percentage of area occupied by p-tau stain). (F) The heatmap of the p-tau density across the entire sample, using the same units and colorbar scale as (E). The scale bars in (A) and (F) are both 5 mm, and the scale bars in (B)–(E) are 250 μ m. AD, Alzheimer's disease.

before being mounted to the Ventana Discovery Ultra (Roche Diagnostics, Rotkreuz, Zug, Switzerland) and processed with the Chromomap DAB kit's staining protocols. After the heat-induced epitope retrieval and background blocking steps, the slides underwent 1 h of AT8 (liquid, Invitrogen, 1:100, RRID: AB_223647) and Ab4G8 (liquid, BioLegend, 1:15,000, RRID: AB_2564633) primary antibody incubations and HRP-DAB chromogenic reactions. The tissue sections were then counterstained with hematoxylin.

Upon completion of the protocols, the slides were placed in water and dehydrated to xylene on the Leica Autostainer XL (Leica Biosystems, Danvers, MA, USA). Finally, each slide was coverslipped by hand with the Leica Surgipath Micromount and imaged with the Zeiss Axio Scan.Z1 (The Zeiss Group, Oberkochen, Germany) with a 20 $\times$ magnification objective. The images were digitized with the Zeiss Zen Blue Microscopy Software (version 3.9) (Figures S2 and S3).

Each tissue section was 30 μ m thick, which was at the upper limit of the autostainer's capabilities. This resulted in damage to two control tissue sections, from the same subject, which were deemed unusable. Due to logistical constraints, additional sections were not stained. Another control tissue volume and AD tissue volume did not have any tissue sections retained after sOCT imaging. In total, there were four AD and three control tissue volumes used for histopathological assessment (with three sections for A β and three sections for p-tau).

The imaging analysis software QuPath²⁶ was used to extract the stain vector from each image (Figure 2A,B), facilitating the identification of the individual A β plaques and p-tau tangles. This process relied upon deconvolution, which separated each pixel into its constituent components based upon user-defined color vectors (Figure 2C).²⁷ The deconvolved stain was then downsampled from an isotropic pixel

size of 0.2195 μ m to 12 μ m, to match the x-y voxel dimensions of the OCT volumes. In addition, a tissue mask was generated for each histopathology image (using Fiji) to exclude the background signal.

For each stain image, the plaques were segmented by applying a minimum intensity threshold. The result was a binarized image, where one corresponded to pathology and zero corresponded to background signal (Figure 2D). A heatmap was generated for each histopathology image by following the same procedure in Section 2.3 (Figure 2E,F). The values of the histopathology heatmap ranged from 0 to 1, where 0 indicated an ROI that was entirely devoid of pathology and 1 indicated an ROI with pathology present in each pixel.

2.5 | Correlation between vascular metrics and histology

We generated a vascular sub-volume for each stained tissue section by cropping the respective OCT volume at the same depth as the stained section. As the tissue sections were collected sequentially during sOCT imaging, we knew the physical depth of each stained tissue section within its respective OCT volume. Each cropped vascular sub-volume had a thickness of 1 mm. Using the uniform sampling method from Section 2.3, a vascular metric heatmap was generated for each vascular OCT sub-volume.

A two-dimensional tissue mask was generated for each of these OCT sub-volumes, following the procedure in Section 2.3. Then, the tissue mask of the histopathology section was registered to the respective OCT sub-volume tissue mask. For the histopathology sections with minimal physical distortion, we used an automated nonlinear

registration tool (Demons Algorithm).^{28–30} The sections with significant distortion required a manual landmark tool that utilized the Insight Segmentation and Registration Toolkit (ITK) from the National Library of Medicine.^{31,32} In addition, masks for the GM and WM were generated for each pair of images. This facilitated analyses of the correlation within the GM and WM, respectively.

Only the vascular metric heatmap ROIs that lay entirely within the GM tissue mask were included in the subsequent analyses. Since the histopathology was coregistered to the OCT sub-volume, each ROI provided a direct mapping between vasculature and pathology. A matrix of vasculature–pathology pair values was generated for each vascular metric and pathology sample.

2.6 | Statistical tests

Most cerebral microvasculature imaging studies report the average value within a distinct anatomical region (e.g., WM, GM, sulci, gyri). This approach requires stereological sampling to achieve adequate power though, which can be time-consuming and expensive. Instead, sOCT produces continuous vascular reconstructions across the entire tissue volume, facilitating analysis of the entire vascular network and removing the need for stereology. This also allows for multiple types of statistical tests.

First, the Kruskal–Wallis test was used to compare the average value of each vascular metric within each subdivision. The Kruskal–Wallis test was followed by Dunnett's multiple comparisons procedure to account for type I errors. Since this method takes the average of each subdivision, it is sensitive to homogeneous changes. One drawback is that it condenses the rich volumetric information into a single value for each subdivision.

To complement this approach, we also compared the distributions of each vascular metric by dividing each tissue volume into uniform ROIs and then measuring the vascular metric within each ROI, creating a three-dimensional metric map. Because each metric map contained multiple measurements from the same subject, these measurements were not independent. This motivated the implementation of the general linear mixed effects (GLME) model, which accounts for correlation between ROIs from the same tissue volume while measuring statistical differences between groups. The GLME model was defined so that the response variable was the vascular metric, the fixed effect was the group (AD or control), and the random effect was the subject identifier (random intercept model). The GLME model used the t-statistic to assess significance.

Both the Kruskal–Wallis and the GLME model statistical tests were used to assess significant differences across the entire tissue volume and each subdivision. Each vascular metric was tested separately for statistical significance. Combined, these two approaches provided insight into localized changes and inter-subject variability for each vascular metric. Due to the relatively small sample size, the Kruskal–Wallis test only detects significance when both the group means are sufficiently disparate and the inter-subject variability is low.

In contrast, the GLME model may detect significant differences when the group means are sufficiently disparate, but the inter-subject variability is relatively high. This can lead to a situation where the Kruskal–Wallis test is insignificant while the GLME model is significant, which would indicate localized alterations and a high degree of inter-subject variability. On the other hand, a significant result for both the GLME model and the Kruskal–Wallis test indicates a uniform change to the vascular morphology across subjects. Overall, the GLME model provides increased sensitivity to localized cerebrovascular changes that would otherwise be averaged out in the Kruskal–Wallis test, which can result in discrepancies between the two statistical test results. Therefore, including both the GLME model and the Kruskal–Wallis test allowed us to assess both the magnitude and distribution of changes to the vascular metrics within each subdivision.

Finally, the Spearman's rank correlation coefficient (ρ) was calculated for each matrix of vasculature–pathology pair values to assess whether the vascular metrics changed with respect to changes in pathology. The Spearman's rank correlation coefficient was performed for each subject, both for $A\beta$ and p-tau.

In addition, we measured the Spearman's rank correlation coefficient across all samples from both groups. For each subject, we calculated the average ROI value for each histopathology density and each vascular metric. Then, the averages from all AD and control samples were combined into a single matrix. This process was repeated separately for the WM and GM. Additionally, we compared the average pathology in the GM with the average vascular metric in the WM. The Spearman's rank correlation coefficient was calculated for each of these groups of averages. The goal was to identify relationships between the severity of pathologies and the vascular alterations, independent of disease states. The scripting language MATLAB (MathWorks Inc., Natick, MA, USA)³³ was used for all statistical analyses, and an α level of 0.05 was used for all tests.

3 | RESULTS

3.1 | Vascular heatmap distributions

The heatmaps visually demonstrated several patterns that reflect alterations within the vascular networks. First, the volume fraction, length density, and branch density heatmaps were highly correlated within the same subject (Figure 3A–C). Additionally, each heatmap contained regions of hypervascularity and hypovascularity. For both length and branch density, the hypervascularity occurred primarily in the gyri, whereas the hypovascularity was predominantly localized in the sulci and deep WM. These hypovascular regions were also extensively connected to one another. Overall, there was still a high degree of heterogeneity both within subjects and across groups. The heatmaps also demonstrated an overall trend of reduced vascularity in AD. In comparison, the tortuosity heatmaps (Figure 3D) did not appear spatially correlated with the other vascular metrics.

All of the heatmap ROIs were combined within each group and then represented as a violin plot (Figure 4) using an open-source MATLAB

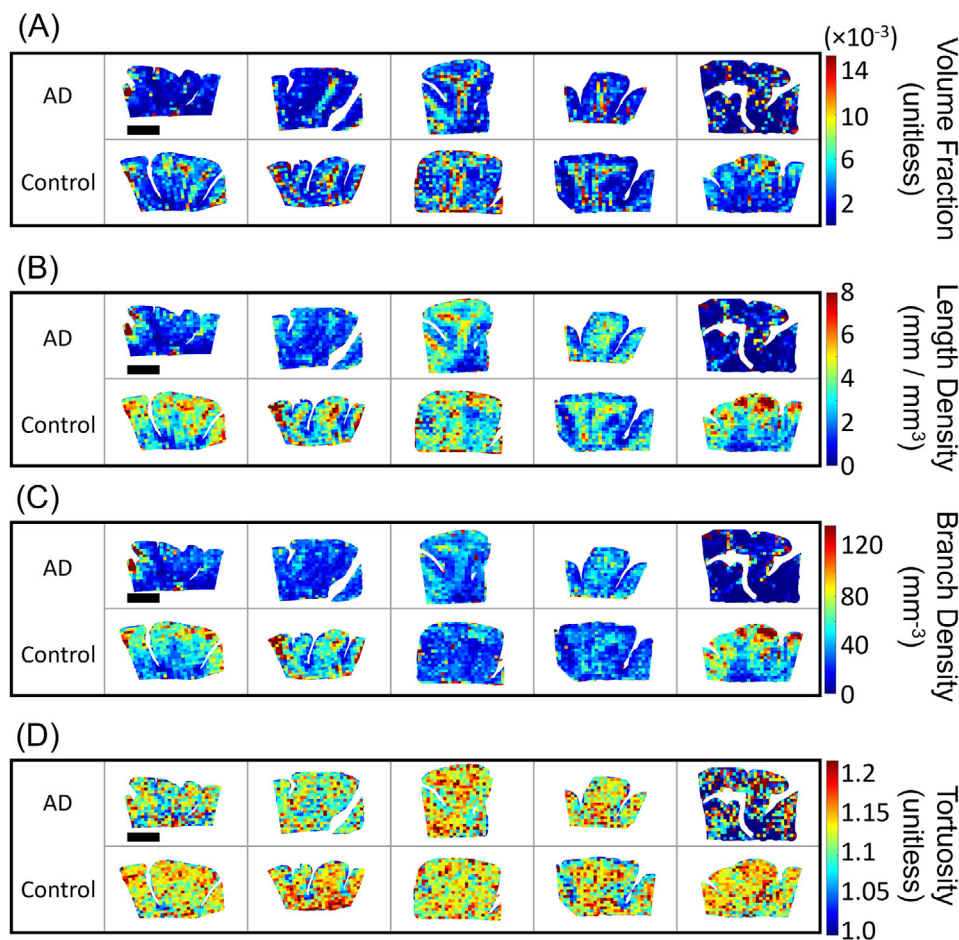


FIGURE 3 Heatmaps for (A) volume fraction, (B) length density, (C) branch density, and (D) tortuosity. The scale bar (in the top-left box of each subfigure) represents 1 cm, and all heatmaps are displayed on the same scale.

package.³⁴ Among the statistically significant violin plots (indicated by an asterisk), AD generally had higher variance than control. This indicates a higher degree of heterogeneity in vasculature across the AD subjects. The shapes of the violin plots provided additional information as well. Most of the violin plots were asymmetric, indicating non-normal distributions for the heatmap ROIs. In addition, the lower portion of the violin plots was wider in the disease groups, indicating a greater proportion of hypovascularity in AD.

The GLME model detected significant differences between AD and control when comparing the distributions of the heatmap ROIs (Figure 4 and Table 1). First, AD demonstrated a reduction in volume fraction across the entire tissue volume ($p < 0.01$), gyri ($p < 0.001$), sulci ($p < 0.05$), GM ($p < 0.01$), WM ($p < 0.05$), and GM gyri ($p < 0.01$). Second, the length density was lower in AD for the entire tissue volume ($p < 0.001$), gyri ($p < 0.001$), sulci ($p < 0.001$), GM ($p < 0.001$), WM ($p < 0.01$), GM sulci ($p < 0.05$), WM sulci ($p < 0.01$), GM gyri ($p < 0.01$), and WM gyri ($p < 0.01$). Third, the branch density was significantly lower in AD across the entire tissue volume ($p < 0.05$), sulci ($p < 0.01$), GM ($p < 0.05$), WM ($p < 0.05$), WM sulci ($p < 0.01$), GM gyri ($p < 0.05$), and WM gyri ($p < 0.05$). There were no significant differences in tortuosity between AD and control.

These quantitative differences were qualitatively apparent in the heatmaps as well (Figure 3). The AD heatmaps demonstrated an overall reduction in volume fraction, length density, and branch density compared to controls. The AD heatmaps also demonstrated the regional deficits in length density that the GLME model detected. The reduced branch density in the sulcal WM was less apparent in the heatmaps. Albeit this may be a result of averaging the heatmap across the entire depth. Overall, the heatmaps facilitate visual discrimination of intersubject and intergroup variability.

3.2 | Average vascular metric

The Kruskal-Wallis test identified significant differences between AD and controls when comparing the mean values in each subdivision (Figure S4). In particular, the volume fraction was lower in the GM ($p < 0.05$) and GM gyri ($p < 0.05$), and the length density was lower in the GM gyri ($p < 0.05$). In addition, the AD group trended lower than controls across all vascular metrics (Figure S4). This mirrored the results in the violin plots from the heatmap ROIs (Figure 4).

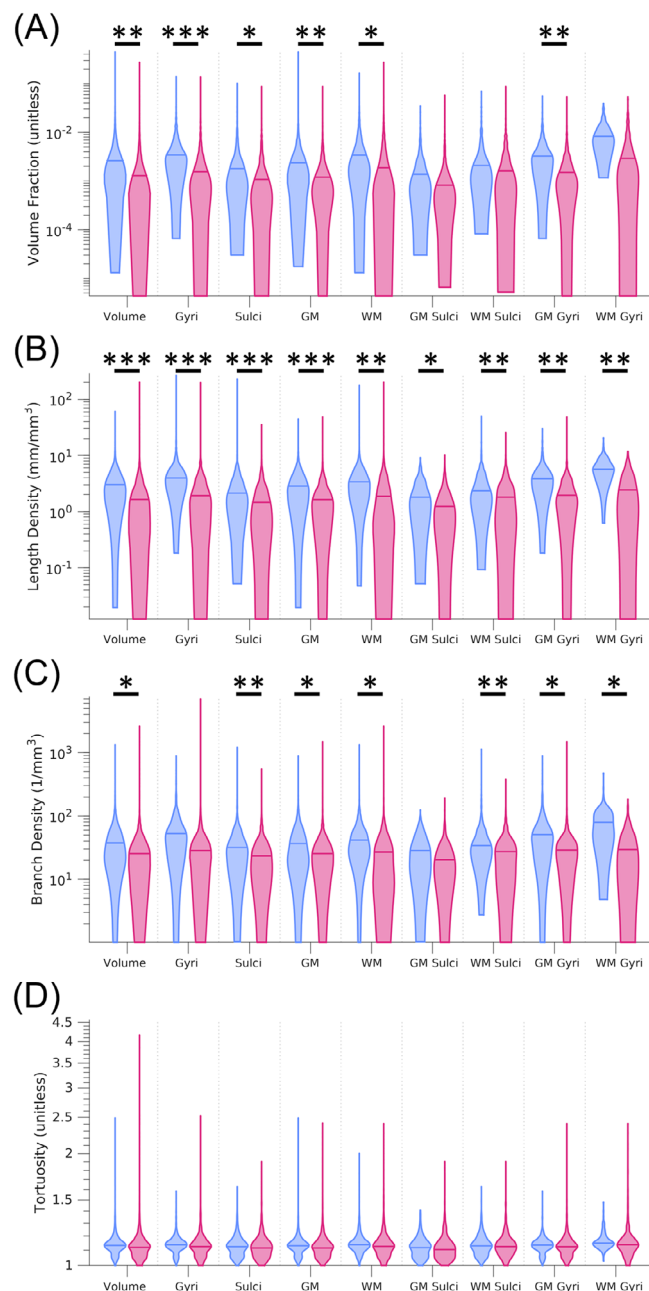


FIGURE 4 Violin plots depicting the distributions of the isotropic ROIs from the three-dimensional tissue volume heatmaps. Each violin plot consists of all ROIs from all subjects within the respective group. Blue represents the control group and red represents the AD group. (A) Volume fraction, (B) length density, (C) branch density, and (D) tortuosity. The brackets and asterisks indicate the statistically significant results from the linear mixed effects model. * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$. AD, Alzheimer's disease; ROI, region of interest.

3.3 | Quantitative pathology

The histopathology heatmaps demonstrated a range of mild to severe density of A β (Figure 5A) and p-tau (Figure 5B). All AD samples contained a low to moderate distribution of A β throughout the GM, yet

only sparse A β plaques throughout the WM. All but one AD sample contained dense distributions of p-tau throughout both the sulcal and gyral GM. All control samples contained relatively few A β plaques and p-tau tangles.

3.4 | Vascular correlation with pathology

The subject-level spatial relationship of pathology and vasculature was convoluted and varied between subjects (Table S2). Among the significant results ($p < 0.05$), the Spearman's rank correlation coefficients were primarily weak ($|r| \leq 0.30$) or moderate ($0.30 < |r| \leq 0.5$) with both negative and positive correlation coefficients between the vascular metrics and the pathologies (A β and p-tau). Despite the heterogeneity across subjects, it was interesting to observe that within subjects, the significant correlations shared the same trend (either all positive or all negative) across all metric pairs.

The AD samples revealed several moderate correlations and one strong correlation ($|r| > 0.5$) within the GM. In particular, the AD subject with the most severe pathology (both for A β and p-tau) demonstrated significantly negative correlations between each pathology and vascular metric (except for p-tau and tortuosity) (Figure 6 and Table S2), and the correlations were primarily moderate and strong. In other samples with weaker pathology (either in A β or p-tau), the correlations were mostly weak and could be either positive or negative. Similarly, the correlations within the WM were primarily weak, likely due to the relatively modest density of pathologies.

While examining the global correlation between pathology and vasculature across subjects, we found several strong correlations. Each point in the scatterplots of Figure 7 represents the subject average from the vascular metric (y-axis) and the pathology (x-axis) within the GM heatmap ROIs. The Spearman's rank correlation coefficient was measured on all the samples (both AD and control) to identify trends between pathological severity and vascular metrics. There was one significant, strong negative correlation between the density of p-tau and branch density ($p = 0.066$, $\rho = -0.75$). Although there was only one significant correlation, all other correlations were negative. In particular, the correlations between volume fraction and both A β and p-tau were strongly negative. When examining the correlations between the density of pathology in the GM with the vascular metrics in the WM, none of the correlations were significant, yet all were negative (Figure S5).

4 | DISCUSSION

4.1 | Vascular reduction in the DLPFC in AD: heatmap distributions

When comparing AD to control, the Kruskal-Wallis test detected significantly lower volume fraction within the GM and the GM gyri. In addition, it also detected reduced length density in the GM gyri. These results indicate that AD demonstrated an overall reduction

TABLE 1 Summary statistics from the significant outcomes of the GLME model for AD vs. control

Vascular metric	Region	p	Coefficient estimate	SE	CI (95%)	df	t	μ_{AD}	SD_{AD}	$\mu_{Control}$	$SD_{Control}$
Volume fraction	Volume	0.00688	0.00180	6.80E-04	[0.00051, 0.0032]	13899	2.7	0.0030	0.0074	0.0050	0.010
	Gyri	0.00002	0.00140	3.27E-04	[0.00076, 0.0020]	2792	4.3	0.00341	0.00821	0.00482	0.0079
	Sulci	0.03179	0.00164	7.65E-04	[0.00014, 0.0031]	2222	2.15	0.00243	0.00594	0.00403	0.0082
	GM	0.00229	0.00185	6.05E-04	[0.00066, 0.003]	8651	3.05	0.00222	0.00436	0.00425	0.0105
	WM	0.04175	0.00211	1.03E-03	[0.000079, 0.0041]	6466	2.04	0.00402	0.00955	0.00615	0.0089
	GM gyri	0.00756	0.00177	6.63E-04	[0.00047, 0.0031]	1870	2.67	0.00284	0.00466	0.00477	0.0055
Length density	Volume	0.00073	1.50	0.440	[0.63, 2.38]	13899	3.38	1.97	3.92	3.54	3.05
	Gyri	1.66E-11	2.00	0.296	[1.42, 2.58]	2792	6.76	2.26	5.85	4.26	9.72
	Sulci	1.73E-08	1.34	0.237	[0.87, 1.82]	2222	5.66	1.57	2.18	2.91	8.31
	GM	0.00067	1.57	0.462	[0.67, 2.48]	8651	3.4	1.81	2.07	3.39	2.85
	WM	0.00247	1.76	0.581	[0.62, 2.9]	6466	3.03	2.11	5.13	3.96	4.91
	GM gyri	0.00166	1.90	0.602	[0.72, 3.1]	1870	3.15	2.28	2.36	4.35	3.28
	GM sulci	0.03650	0.769	0.367	[0.048, 1.49]	1143	2.09	1.28	1.36	2.16	1.60
	WM gyri	0.00919	2.98	1.141	[0.74, 5.2]	549	2.61	2.49	2.43	5.97	3.16
	WM sulci	0.00634	1.67	0.612	[0.47, 2.9]	1216	2.73	1.79	2.09	3.11	3.70
Branch density	Volume	0.01700	20.2	8.40	[3.7, 36.7]	13899	2.4	27.9	56.9	47.9	52.3
	Sulci	0.00202	18.6	6.02	[6.8, 30.4]	2222	3.09	21.9	28.5	39.7	70.2
	GM	0.02054	22.3	9.64	[3.44, 41.2]	8651	2.32	26.9	43.9	47.4	48.6
	WM	0.02529	21.2	9.47	[2.62, 39.8]	6466	2.24	28.0	78.3	51.3	66.7
	GM gyri	0.03772	27.7	13.3	[1.57, 54]	1870	2.08	32.7	53.6	69.6	78.9
	GM sulci	0.06324	10.6	5.69	[-0.58, 21.8]	1143	1.86	19.2	19.1	31.5	21.7
	WM gyri	0.02728	37.8	17.1	[4.2, 71]	549	2.21	30.4	29.8	79.0	59.2
	WM sulci	0.00196	25.5	8.21	[9.4, 42]	1216	3.1	24.3	28.2	46.9	75.6

Abbreviations: CI, confidence interval; df, degrees of freedom; p, p-value; SD, standard deviation; SE, standard error of the coefficient estimate; t, t-statistic; μ , mean.

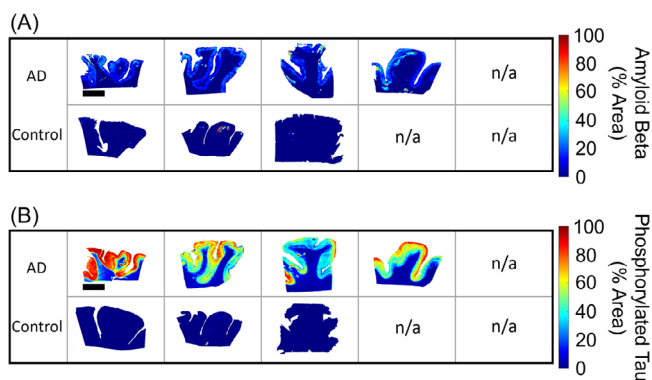


FIGURE 5 Histopathology heatmaps for (A) Aβ and (B) p-tau. The units of the heatmap represent the percentage of the region of interest containing pathology. The scale bar (in the top-left box of each subfigure) represents 1 cm, and all heatmaps are displayed on the same scale. Aβ, amyloid β; p-tau, phosphorylated tau.

in GM mesoscopic vascularity, coinciding with a prior study showing disrupted cortical microvascular integrity in AD.³⁵

The GLME model identified numerous significant differences across subdivisions that were insignificant with the Kruskal-Wallis test. This discrepancy indicates intra- and inter-subject heterogeneity. This coincides with capillary changes in AD, which tend to present with non-uniform distributions,^{36,37} indicating that this heterogeneous pattern may extend up from the capillaries to the mesoscopic vessels.

First, the GLME model identified several subdivisions with significantly reduced volume fraction. In particular, the strongest significance was within the gyri (including both the GM and WM). Since penetrating arterioles supply oxygenated blood to the gyri,³⁸ the deficits to cortical vasculature in AD may lead to downstream degradations in the WM gyral vasculature, leading to an overall reduction in volume fraction.

Second, the GLME model detected significantly reduced length density across the overall volume and every subdivision, making length density the most sensitive metric for cerebrovascular alterations in AD. The strongest significance was seen within the overall volume, gyri, sulci, and GM. A relevant study identified reduced capillary length density across most neocortical areas of AD patients.³⁹ Our result shows that these changes occur across non-cortical regions as well. Significant reductions to the length density have also been reported for capillaries

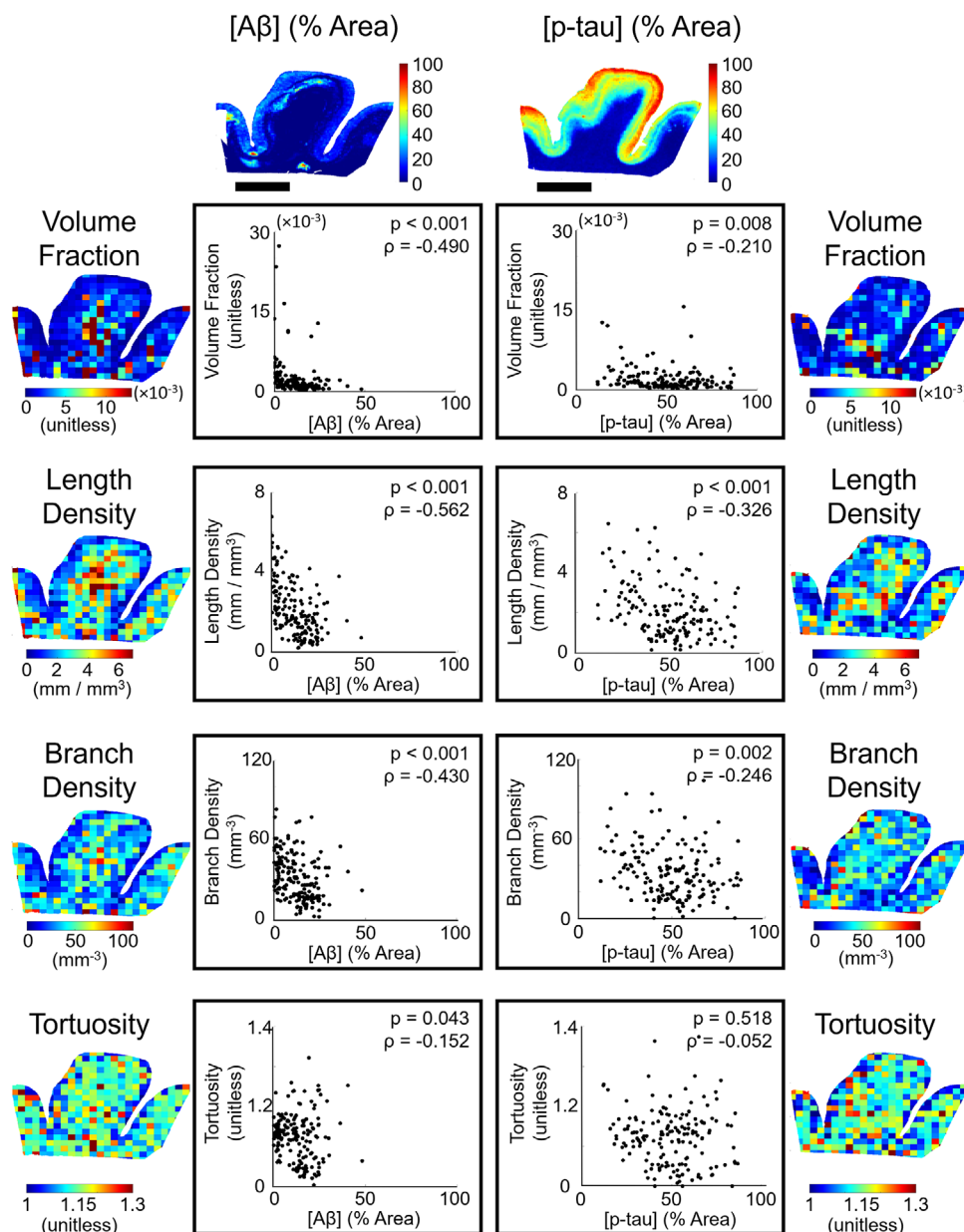


FIGURE 6 Comparison of the histopathology heatmaps of Aβ (top left) and p-tau (top right) with the respective vascular heatmaps (far-left and far-right columns, respectively). The vascular heatmaps are a 1 cm z-axis projection centered about the same depth as the respective histopathology heatmap. The far-left and far-right heatmaps are co-registered with the Aβ and p-tau heatmaps, respectively. Each row of the vascular heatmaps represents one of the vascular metrics (from top to bottom: volume fraction [unitless], length density [mm / mm³], branch density [mm⁻³], and tortuosity [tortuosity]). All heatmaps are from the same subject, and differences between the histopathology and vascular heatmaps are due to tissue distortion from mounting the sample onto a slide. The scatter plots in the middle two columns depict the co-registered regions of interest between the vascular metric (y-axis) and the concentration of pathology (x-axis). The scatter plots in the center-left column compare the respective vascular heatmap in the left column against the Aβ heatmap. The scatter plots in the center-right column compare the respective vascular heatmap in the right column against the p-tau heatmap. The scale bar represents 1 cm, and all heatmaps are displayed on the same scale. The values of p and ρ in each scatterplot represent the values from the Spearman's rank correlation coefficient. Aβ, amyloid β; p-tau, phosphorylated tau.

within human WM.³⁶ Therefore, these results, and prior studies, suggest that reductions to WM length density may occur across a range of vessel sizes in AD.

Additionally, the most significant reductions in branch density were in the sulci and WM sulci in AD. One mechanism may be linked to

the overall reduction in vasculature of the WM. The reduced length density in the WM vessels may manifest as vascular fragmentation, appearing as decreased branching density, which has been noted in the cortical capillary bed of AD patients.³⁹ Alternatively, the sulcal WM branch density reduction may have been a secondary response to

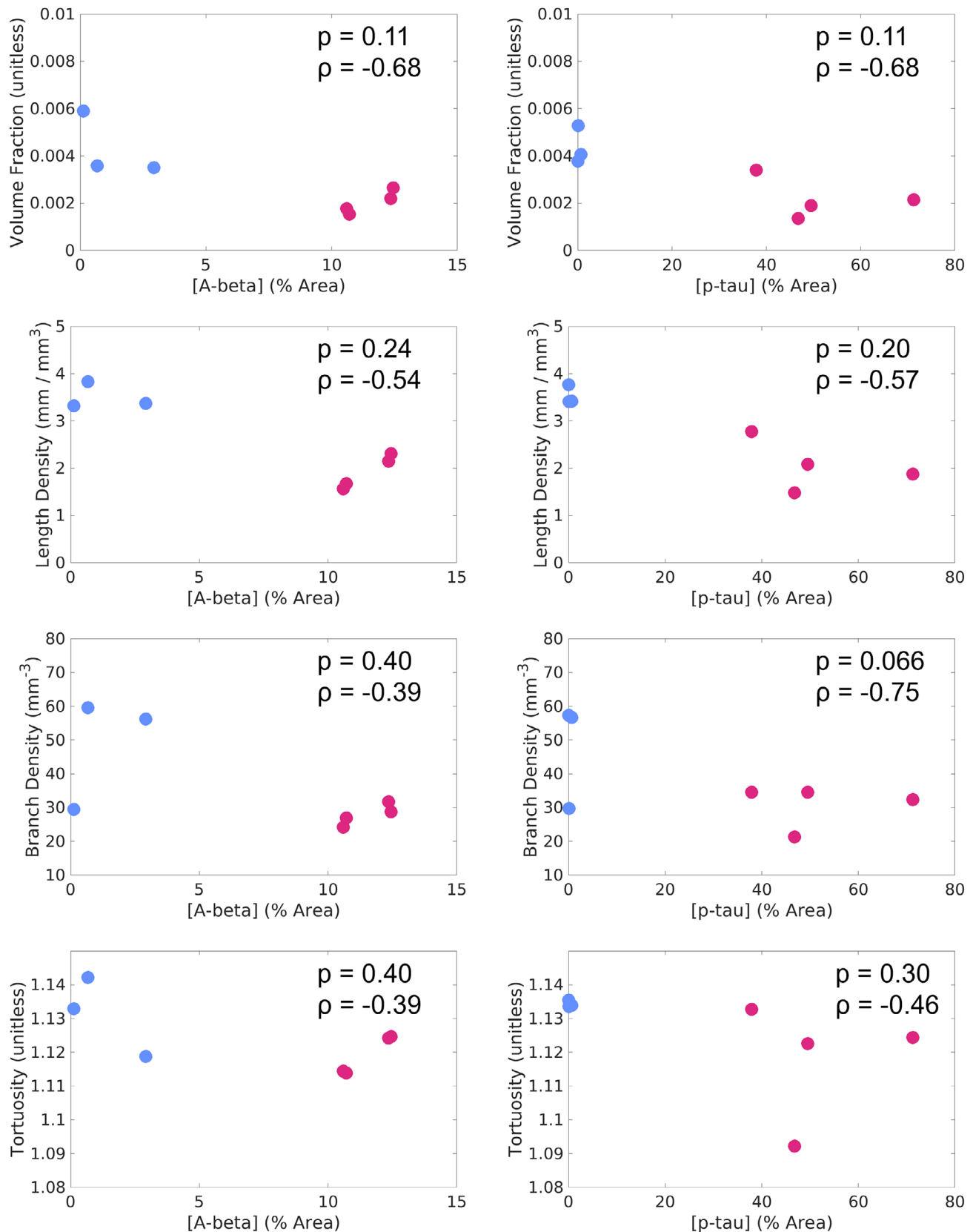


FIGURE 7 Scatter plots of the mean GM vascular metric (y-axis) versus the mean GM percentage area of histopathology for each subject (x-axis). Each point represents the means of a single subject. Each mean is the average of all GM heatmap ROIs within a single subject. Blue and red represent the control group and the AD group, respectively. The p and ρ represent the p -value and rho, respectively, from the Spearman's rank correlation coefficient. AD, Alzheimer's disease; GM, gray matter; ROI, region of interest.

neurodegeneration. In AD, there is a significant accumulation of NFTs in the prefrontal cortex,⁴⁰ which, in turn, correlates with a loss of vulnerable neurons.^{41,42} This progressive process of NFT aggregation and neuronal death leads to the marked cortical atrophy typically observed in AD.⁴³ This may result in fragmentation of the local vasculature, which can manifest as reduced branch density and other morphological alterations of the blood vessels.³⁹ Overall, reduced WM vascularity in the vicinity of the cerebral cortex could be a robust marker for detecting abnormalities in AD.

4.2 | Histopathology and vasculature

Generally, the A β was uniformly distributed across the GM of all AD subjects, while the p-tau ranged from moderate to severe. A complex pattern was revealed in the intrasubject correlations between pathological burden and vascular metrics in the GM. The AD subject with severe pathology demonstrated negative correlations between GM pathologies and vascular metrics. In contrast, moderate pathology subjects exhibited a combination of weakly positive or negative correlations. This suggests an initial phase of mixed vascular alterations in response to mild pathology, followed by a marked decline in vascularity coinciding with severe pathology. Prior studies on the impact of A β on capillaries have demonstrated both hyper- and hypo-vascularization, and the phenomenon may extend to mesoscopic vessels.^{44,45} Likewise, prior work has shown a link between a high pathological burden in AD and increased cerebrovascular inflammatory molecules, which could lead to vascular degeneration.⁴⁶

Combining the subject averages across both groups enabled an analysis over a wider range of pathological burden. Despite the lack of significant correlations, there were negative relationships between A β and p-tau and each vascular metric. This may indicate that increased pathological burden correlates with reduced mesoscopic vascularity in the GM. These findings are consistent with prior studies that identified a correlation between A β burden and cortical vascular dysfunction in AD.⁴⁷

4.3 | Strengths, limitations, and future directions

sOCT enables reconstruction of continuous, mesoscopic vascular networks without histochemical labeling by leveraging endogenous sources of contrast in tissue—discontinuities in the refractive indices and the related differences in the scattering coefficients. This facilitates label-free vascular imaging and circumvents staining and mounting artifacts. Moreover, sOCT is nearly distortion-free, producing a morphologically accurate reconstruction of the vascular network. sOCT can also resolve vessels that are too small for in vivo imaging systems like magnetic resonance imaging and computed tomography. Overall, these advantages make sOCT uniquely suited for label-free, mesoscopic vascular imaging of volumes on the order of cubic centimeters.

One major limitation of traditional histology is that variations in transverse sampling of the cortical ribbon can lead to a sampling bias.

sOCT overcomes this limitation through comprehensive volumetric imaging. Since each tissue volume contained roughly equal proportions of each neocortical layer, sOCT enabled volumetric analyses of the continuous vascular network across all neocortical layers. Therefore, small variations in transverse sectioning were averaged out across subjects. Although the cross-section of the tissue volume in Figure S1 may appear non-uniform in the transverse axis, the quantitative analyses included the vasculature across all cortical layers of the three-dimensional volume.

While our study was novel in examining mesoscopic arterioles and venules, it had several limitations. The literature on cerebrovasculature focuses predominantly on arteries, veins, and capillaries. Since we only examined the mesoscopic vasculature, we were unable to directly cross-validate our outcomes. We also lacked groups of pure AD without CAA, due in part to constraints of sample availability. CAA is a common comorbidity in AD, with a recent histology meta-analysis identifying moderate-to-severe CAA among 48% of AD donors.⁴⁸ Although CAA leads to cerebrovascular dysfunction,⁴⁹ which could be a confounding factor of the observed vascular alterations, further histopathological examination of the DLPFC samples in this study identified very mild, sparse CAA. This is in accordance with the literature, since the distribution of CAA pathology is highly variable (both within and across brain regions)⁵⁰ and late-stage AD typically presents with a greater distribution of CAA in the occipital lobe.⁵¹ Therefore, the majority of vascular morphology differences between the AD and control groups were associated with the AD pathology instead of CAA.

This study examined late-stage AD, which primarily occurs in old age when cerebrovascular disease (CVD) becomes a comorbidity.⁵² Consequently, all AD and age-matched control samples had arteriosclerosis—a form of CVD that results in arterial stiffening, which can extend to the arterioles.⁵³ Since all controls also had arteriosclerosis, the significant differences between the groups are attributed primarily to AD rather than a combination of AD and CVD. There were minor discrepancies among groups, though, when controlling arteriosclerosis disease staging. First, the arteriosclerosis severity was slightly lower in the control group than in AD (Table S1). The control group had two mild, two moderate, and one severe case, whereas the AD group had three moderate and two severe cases. The average arteriosclerosis stage was slightly higher in AD, which may have been a moderating factor, potentially increasing the vascular morphology differences between groups. Additionally, one of the control subjects was Braak stage IV whereas all other controls were Braak stage 0. Given that all AD subjects were Braak stage VI, and AD demonstrated a pronounced reduction in vascularity, a control group composed entirely of Braak stage 0 likely would have increased the statistical significance between groups.

Additionally, the AD and control groups were not sex-matched, which may have introduced a potentially confounding variable. The UNITE Brain Bank specializes in chronic traumatic encephalopathy (CTE) and primarily samples male donors, which led to the all-male control group, resulting in an unbalanced sex between AD and controls. Three of the controls had a history of repetitive head impacts (RHI) (without CTE) from American football, while two had no history of contact sports nor RHI. Since RHI can alter both cerebrovasculature

and WM pathologies,⁵⁴ it might introduce a potentially confounding variable. Future studies will include a more general comparison group—both sex-matched and devoid of RHI.

Future directions will also include additional anatomical locations beyond the DLPFC, which undergoes early A β and late p-tau progression.^{12,13} To identify comprehensive changes to mesoscopic cerebrovasculature in AD and their correlation with A β and p-tau, we will expand the study to regions like the temporal lobe—where early p-tau is commonly reported.¹² Additionally, incorporating a CAA cohort without AD and imaging the occipital lobe—where CAA pathology most frequently occurs⁵⁵—would further strengthen our outcomes.

In conclusion, we demonstrated the value of this novel application of sOCT in the investigation of mesoscopic vascular alterations in AD. sOCT enabled high-throughput, label-free cerebrovascular imaging and facilitated cross-correlation with A β plaques and NFTs via traditional histopathology. Our method opens new avenues to examine this spatial correlation across unprecedented scales—with great potential to elucidate a comprehensive understanding of the role of vascular alterations in neurodegenerative disorders.

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CONFLICT OF INTEREST STATEMENT

None of the authors have any conflicts of interest within the scope of this study. Author disclosures are available in the [Supporting Information](#).

CONSENT STATEMENT

Prior to donation, all AD subjects were consented through the Health Outreach Program for the Elderly (HOPE) study—part of Boston University's Alzheimer's Disease Research Center (ADRC). All control subjects provided consent for brain donation through the Understanding Neurological Injury and Traumatic Encephalopathy (UNITE) Brain Bank, prior to donation. The tissue samples were collected *post mortem*, anonymized, and then transferred in accordance with institutional protocols. According to the Department of Health and Human Services (45 CFR 46), these samples were no longer bound by human subject regulations, since the donors had already consented to research and the samples were anonymized. Therefore, this study did not require additional consent from the donors. Regardless, this study followed all ethical guidelines across Boston University, Boston University's ADRC, and the UNITE Brain Bank.

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